Notes: Christoffersen, Evans, and Musto (JF, 2013)

What Do Consumers' Fund Flows Maximize? Evidence from Their Brokers' Incentives

Contents

1 Big picture

- **Question.** Do mutual fund flows reflect investors' own preferences and fund quality, or do they reflect the compensation incentives of the brokers who intermediate the flows?
- **Setting.** The paper studies U.S. mutual funds from 1993 to 2009 by merging SEC N-SAR filings with Morningstar data. N-SAR filings are central because they disclose actual broker payments and separate inflows from outflows.
- **Main incentive variables.** The paper focuses on (i) the portion of front-end sales loads paid to brokers, (ii) whether the broker is captive or unaffiliated, and (iii) revenue-sharing payments or unreimbursed 12b-1 payments made to brokers.
- **Core flow result.** Funds attract more inflows when they pay brokers more, especially through unaffiliated brokers. In contrast, higher total loads paid by investors reduce inflows.
- **Broker channel asymmetry.** The effect of broker payments on inflows is strongest for unaffiliated brokers because unaffiliated brokers sell funds from multiple families and face stronger competition for shelf space and attention.
- **Revenue-sharing result.** Excess revenue sharing also predicts higher inflows in the subset of funds using defensive 12b-1 plans, but its performance consequences differ from load sharing.
- **Performance result.** Funds that pay more of the load to brokers subsequently underperform. The underperformance is strongest for load payments to unaffiliated brokers.
- **Important distinction.** Load sharing predicts poor future performance, but revenue sharing does not once load sharing is controlled for. This difference is consistent with the fact that revenue sharing gives brokers continuing exposure to the fund's asset base, while front-end load sharing rewards only the initial sale.
- **Captive brokerage result.** Captive brokerage has one apparent advantage: it helps fund families recapture redemptions elsewhere in the same family.
- **Policy takeaway.** Broker compensation affects consumer fund choices and can reduce investor welfare. Disclosure of cross-sectional broker payments and the nature of broker compensation is therefore important for consumer protection and fiduciary-duty debates.

2 Incremental contribution of the paper

2.1 Contribution relative to the mutual fund flow literature

Earlier mutual fund flow papers show that flows respond to past performance, fees, loads, and fund family effects. However, most previous papers observe only fund-level loads or net flows. They generally cannot see how much of a sales load is retained by the broker, how much goes to captive versus unaffiliated brokers, or whether flows are new purchases rather than redemptions.

- **Increment 1: actual broker compensation.** The paper moves beyond maximum loads reported in prospectuses and uses actual N-SAR data on dollars paid to brokers.
- **Increment 2: inflows versus outflows.** The paper separates inflows, inflows subject to load, redemptions, reinvestments, and net flows, allowing the authors to test sales incentives directly.
- **Increment 3: captive versus unaffiliated broker channels.** The paper shows that broker incentives have different implications depending on whether brokers are tied to the fund family or sell funds from many families.

Interpretation: The incremental contribution is measurement-based but economically powerful: the paper can observe the incentive contract facing brokers more directly than earlier flow studies.

2.2 Contribution relative to studies of broker-sold funds

Prior evidence shows that broker-sold funds often perform worse than direct-sold funds and that load-paying flows do not appear especially smart. This paper asks a more granular question: within broker-sold funds, does the fund's payment to the broker explain where consumers' money goes and whether those choices work out?

- The paper does not merely compare broker-sold and direct-sold funds.
- It measures how much a fund pays its broker relative to similar funds.
- It then tracks whether the resulting flows predict subsequent performance.

Interpretation: This turns the paper from a distribution-channel comparison into a test of whether broker compensation distorts advice.

2.3 Contribution relative to regulatory debates

The paper directly informs debates over broker fiduciary duties, revenue-sharing disclosure, and conflicts of interest in retail financial advice.

- It studies exactly the payment streams that regulators worry might distort broker recommendations.
- It links those payments to both flow outcomes and consumer performance outcomes.
- It distinguishes upfront sales-load incentives from continuing revenue-sharing incentives.

Interpretation: The paper provides empirical evidence that the current industrial organization of investment advice has real consequences for consumers, while also showing that not all broker payments have the same welfare implications.

2.4 Contribution relative to the investment-advice literature

The paper contributes to the broader literature on delegated retail advice by showing that the advice problem is not just one of investor sophistication or fund quality. It is also a problem of adviser compensation design.

- The consumer delegates search and recommendation to a broker.
- The fund sponsor can alter the broker's incentives through load sharing and revenue sharing.
- The observed allocation of consumer dollars responds to the broker's compensation rather than only to fund performance or investor costs.

Interpretation: The paper's distinctive insight is that broker incentives are embedded in the product-distribution system and are empirically visible in fund flows.

3 Institutional background and hypotheses

3.1 Regulatory background

The SEC has been concerned about broker incentives in mutual fund sales since the early decades of the mutual fund industry. The paper emphasizes several historical points:

- Early SEC and congressional studies worried that mutual fund sales incentives might emphasize asset growth rather than investor welfare.
- Form N-1R, the predecessor to N-SAR, required information about commissions charged and retained by underwriters and dealers.
- Form N-SAR eventually standardized the data needed to measure actual loads, broker shares, and the inflows associated with these payments.
- Modern enforcement actions and FINRA proposals about revenue-sharing disclosure reflect the same concern: compensation may bias broker recommendations.

Interpretation: The paper is not simply motivated by academic curiosity. It answers a long-standing regulatory question using the disclosure regime created for that purpose.

3.2 Mutual fund structure

A mutual fund contracts with several service providers:

- an investment adviser that manages the portfolio;
- a principal underwriter that distributes the shares;
- a transfer agent, custodian, administrator, and other service providers.

The key distinction for the paper is between two payment contracts:

- **Load sharing:** the principal underwriter shares part of the front-end sales load with captive or unaffiliated sales forces.
- **Revenue sharing:** the investment adviser pays brokers out of adviser revenue or related 12b-1 arrangements.

Interpretation: Load sharing is tied to the initial sale; revenue sharing is more closely tied to the continuing asset base. This difference matters for performance incentives.

3.3 Captive versus unaffiliated brokers

- **Captive brokers** sell primarily the funds represented by the sponsor or affiliated underwriter.
- **Unaffiliated brokers** have contracts with many fund families and can direct investor money across a larger set of alternatives.

Interpretation: Unaffiliated brokers create more competition across funds and across sponsor payments. Therefore the effect of broker compensation on flows should be stronger in the unaffiliated channel.

3.4 Hypotheses about loads and inflows

The paper's first hypotheses are about load-sharing incentives:

$$\begin{aligned} H1 &: \text{ Inflows are positively related to the load paid to the broker.} \\ H2 &: \text{ Inflows are negatively related to the total load paid by investors.} \end{aligned}$$

The logic is straightforward. Brokers prefer funds that pay them more. Investors, however, should dislike higher total loads because higher loads raise the investor's cost.

3.5 Hypotheses about broker affiliation

The paper further predicts that unaffiliated brokerage magnifies both effects:

$H3$: Inflows are more negatively related to total loads through unaffiliated brokers.

$H4$: Inflows are more positively related to broker payments through unaffiliated brokers.

Interpretation: Unaffiliated brokers compare many funds, so fund families compete harder for their attention. That competition makes broker compensation more consequential.

3.6 Hypotheses about revenue sharing

Revenue-sharing payments are less transparent to the investor than loads, but they can still affect the broker's incentives. The paper predicts:

$H5$: Inflows are positively related to excess revenue sharing.

Interpretation: Revenue sharing resembles shelf-space payments: it can lead brokers or broker-dealers to favor some fund families over others.

3.7 Hypotheses about performance

The paper tests whether broker-induced flows work out for investors:

$H6$: Funds paying more excess load to brokers subsequently underperform.

$H7$: Funds paying more excess revenue sharing may or may not underperform.

Interpretation: Load sharing gives the broker a strong incentive to sell immediately, without direct exposure to future performance. Revenue sharing is paid from the continuing asset base, so it may align brokers more closely with investors' realized outcomes.

4 Data and measurement

4.1 Data sources

The paper merges two main data sources:

- **SEC N-SAR filings:** provide actual fund-level load payments, broker payment allocation, monthly flow components, expenses, 12b-1 information, and family structure.
- **Morningstar:** provides fund returns, categories, maximum loads, share-class information, fund size, and objective classifications.

The sample covers U.S. mutual funds from 1993 to 2009.

Interpretation: N-SAR is the crucial contribution because it reveals payment variables that are not available in standard mutual fund datasets.

4.2 Loads and load sharing

The main load variables are constructed from N-SAR questions:

$$FrontLoad_{i,t} = \frac{\text{total front load paid by investors}}{\text{inflows subject to load}}, \tag{1}$$

$$LoadPaidToBroker_{i,t} = \frac{\text{dollars paid to captive and unaffiliated brokers}}{\text{inflows subject to load}}. \tag{2}$$

The paper also separates:

- load paid to captive brokers;
- load paid to unaffiliated brokers;
- family average load paid to brokers.

Interpretation: These variables measure the actual compensation incentive for brokers, not merely the maximum load a fund could charge.

4.3 Back loads and redemption fees

N-SAR reports dollars from deferred or contingent deferred sales loads. Because it does not report outflows subject to these loads, the paper defines:

$$BackLoad_{i,t} = \frac{\text{dollars from back loads}}{\text{total outflows}}. \tag{3}$$

A redemption-fee indicator captures whether the fund charges a redemption fee separately from deferred sales charges.

Interpretation: Back loads and redemption fees control for other distribution-related charges that can affect flows and redemptions.

4.4 Expenses and revenue sharing

Expenses are annualized from N-SAR total expenses net of reimbursements:

$$Expenses_{i,t} = 2 \times \frac{\text{N-SAR expenses net of reimbursements}}{\text{monthly average net assets}}. \tag{4}$$

Revenue sharing is proxied using defensive 12b-1 or unreimbursed 12b-1 payments to brokers:

$$RevenueSharing_{i,t} = \frac{\text{defensive 12b-1 or unreimbursed payments to brokers}}{\text{average net assets}}. \tag{5}$$

Interpretation: Revenue sharing is observed only for a subset of funds, so the paper first models selection into the revenue-sharing sample before using revenue-sharing residuals in flow and performance regressions.

4.5 Fund flows

N-SAR separates flow components:

- new inflows;
- inflows subject to a load;
- inflows not subject to a load;
- redemptions;
- reinvestments;
- net flows.

Flow variables are generally scaled by total net assets.

Interpretation: Separating inflows from redemptions is essential. Broker incentives should affect new purchases more directly than redemptions.

4.6 Fund performance

Performance is measured using Morningstar category-adjusted returns. Lagged ranked return is the fund's fractional rank within its Morningstar objective category over the prior 12 months.

Future performance is measured as the compounded net return from month $t+1$ through $t+12$ in excess of the fund's Morningstar objective category return.

Interpretation: Category adjustment compares funds to relevant competitors, not to the full universe of mutual funds.

5 Empirical specifications

5.1 Baseline load-payment regression

The first-stage regression estimates normal load paid to brokers:

$$LoadPaidToBroker_{i,t} = \alpha + X'_{i,t}\beta + \gamma_{cat} + \delta_t + u_{i,t}. \tag{6}$$

The residual is:

$$ExcessLoadPaidToBroker_{i,t} = u_{i,t}. \tag{7}$$

Interpretation: Excess load paid to brokers is the broker payment above what is predicted by observable fund and family characteristics.

5.2 Revenue-sharing selection and baseline regression

The paper first estimates a probit for whether the fund is in the revenue-sharing sample. Then, within the revenue-sharing sample, it estimates:

$$RevenueSharing_{i,t} = \alpha + X'_{i,t}\beta + \gamma_{cat} + \delta_t + v_{i,t}. \tag{8}$$

The residual is:

$$ExcessRevenueSharing_{i,t} = v_{i,t}. \tag{9}$$

Interpretation: The two-stage structure controls for the fact that not all funds disclose or use defensive revenue-sharing arrangements.

5.3 Flow regressions with excess broker payments

The main flow regressions use monthly inflows or redemptions scaled by total net assets:

$$Flow_{i,t} = \alpha + \beta_1 ExcessPayment_{i,t} + \beta_2 Fees_{i,t} + \beta_3 Performance_{i,t-1} + \beta_4 Controls_{i,t} + \gamma_{cat} + \delta_t + \epsilon_{i,t}. \tag{10}$$

For load sharing, the key variable is $ExcessLoadPaidToBroker$. For revenue sharing, it is $ExcessRevenueSharing$.

Interpretation: These regressions test whether broker compensation affects the allocation of consumer flows after controlling for performance, costs, category flows, and family effects.

5.4 Affiliation interactions

The paper interacts excess load payments and other variables with broker type:

$$Flow_{i,t} = \alpha + \beta_c(ExcessLoad_{i,t} \times Captive_i) + \beta_u(ExcessLoad_{i,t} \times Unaffiliated_i) \tag{11}$$
$$+ \lambda_c(Load_{i,t} \times Captive_i) + \lambda_u(Load_{i,t} \times Unaffiliated_i) + Controls + \epsilon_{i,t}. \tag{12}$$

Interpretation: Comparing β_c and β_u tests whether broker payments matter more for unaffiliated brokers. Comparing λ_c and λ_u tests whether investor cost sensitivity differs across channels.

5.5 Performance regressions

Future performance regressions take the form:

$$ExcessReturn_{i,t+1:t+12} = \alpha + \beta ExcessPayment_{i,t} + \theta Flow_{i,t} + \phi Controls_{i,t} + \gamma_{cat} + \delta_t + \epsilon_{i,t}. \tag{13}$$

Interpretation: These regressions test whether broker compensation predicts worse investment outcomes for consumers.

6 Identification and interpretation

6.1 What identifies the broker-payment effect

The identifying variation is cross-sectional and time-series variation in broker payments after removing predictable components of payment levels.

- The first-stage residuals isolate payment deviations from fund and family norms.
- Flow regressions control for fund size, family size, expenses, loads, category flows, redemptions, and past performance.
- Year fixed effects absorb aggregate trends in the mutual fund industry.
- Objective category fixed effects absorb differences across investment styles.

Interpretation: The paper does not claim random assignment of broker payments. It shows that flow and performance outcomes are systematically related to excess payments after extensive controls.

6.2 Why captive versus unaffiliated brokers are informative

Unaffiliated brokers are important for identification because they sell funds from multiple families. This makes the fund family's payment to the broker more likely to influence fund choice.

- Captive brokers have fewer competing funds to sell.
- Unaffiliated brokers face a broader menu of possible products.
- If compensation distorts advice, the effect should be strongest in the unaffiliated channel.

Interpretation: The paper's evidence is stronger because the channel with the most competition for broker attention is also the channel where excess broker payments matter most.

6.3 Why performance tests matter

The flow tests show that broker payments affect consumer choices. The performance tests ask whether those choices are good for consumers.

- If brokers are paid more because they provide better advice, higher broker payments might predict better future performance.
- If brokers are paid more because sponsors buy distribution, higher broker payments may predict worse future performance.
- The paper finds the latter for load sharing, especially through unaffiliated brokers.

Interpretation: The performance evidence connects broker incentives to welfare rather than merely to fund sales.

6.4 Limitations

- Broker payments are not randomly assigned.
- N-SAR reports fund-level aggregates, not investor-level broker recommendations.
- Revenue-sharing proxies are indirect and apply to a smaller subsample.
- Advertising, platform access, or other unobserved distribution efforts may correlate with broker payments.
- The paper mitigates these issues with first-stage residuals, family controls, category controls, year effects, and performance tests.

Interpretation: The results are best interpreted as strong evidence of incentive-consistent distortions in broker-intermediated flows, not as a clean randomized experiment.

7 Tables and figures

7.1 Table I: Descriptive statistics

Read:

- The sample includes 268,906 observations for many flow variables.
- Average inflows are 4.22% of TNA, average redemptions are 3.12%, and average net flows are 1.42%.
- Average fund size is about $1.118 billion, and average expenses are 1.17% of TNA.
- Front load paid to brokers averages 1.79% of inflows for captive brokers and 2.30% for unaffiliated brokers.
- Maximum Morningstar front loads are around 4.57% for captive channel funds and 4.64% for unaffiliated channel funds.
- Defensive 12b-1 payments average 0.17% of TNA among funds with defensive 12b-1 plans.

Economic message: Funds pay substantial amounts to brokers, and actual N-SAR broker payments are distinct from maximum loads listed in Morningstar. This is why the paper's data are valuable.

Descriptive Statistics

This table provides descriptive statistics of our matched sample of Morningstar and N-SAR data from 1993 to 2009 and considers only those funds with a 12b-1 plan, a defensive 12b-1 plan, or load. Panel A gives the mean, median, and standard deviation for key variables used in the analysis. The variables *Inflows, Inflows Subject to a Load, Redemptions, Reinvestments* and *Net Flows* are all based on information from Question 28a to 28f in the N-SAR file, where reinvestments are calculated as a percent of assets like inflows and redemptions and net flows are inflows plus reinvestments less redemptions. A fund is defined as captive (unaffiliated) if the front load is paid only to captive (unaffiliated) brokers and no load payment goes to unaffiliated (captive) brokers. Defensive 12b-1 plans and 12b-1 plans are determined as the total dollar value of the plan (Q44 or Q43, respectively) divided by average net assets (Q75b) and multiplied by two to annualize. The load information and its allocation by captive or unaffiliated sales channel is based on information from Q28 to Q38 in the N-SAR file. Front load paid by investors is given as both the maximum amount paid by any one individual in the fund (from Morningstar) and the actual amount paid across all individuals in the fund (from N-SAR). Panel B provides correlation between the actual loads collected that are reported on the N-SAR files and the maximum load charged reported in Morningstar. The conditional correlations condition on the front load and back load being positive as reported in Morningstar, or over 80% of the fund total net asset being in the A or B share class. Additional details on variable construction are provided in Appendix A.

Panel A: Variable Means, Medians, and Standard Deviations

Variables	Units	Mean	Median	*SD*	Obs
New Inflows	($ thousands)	30,064	4,069	129,943	268,906
Redemptions	($ thousands)	25,344	3,974	105,271	268,906
Reinvestments	($ thousands)	4,403	0	72,837	268,906
Net Flows	($ thousands)	9,191	135	102,954	268,906
Inflows	% TNA	4.22%	2.13%	6.11%	268,906
Inflows Subject to Load	% TNA	1.40%	0.48%	2.96%	182,330
Redemptions	% TNA	3.12%	1.96%	4.27%	268,906
Reinvestments	% TNA	0.31%	0	1.39%	268,906
Net Flows	% TNA	1.42%	0.20%	6.09%	268,906
Fund Size	$ millions	1,118	182	5,196	268,906
Expenses	% TNA	1.17%	1.09%	0.58%	237,614
Front Load Paid to Brokers					
Captive	% Inflow	1.73%	1.31%	1.60%	25,807
Unaffiliated	% Inflow	2.30%	2.01%	1.80%	123,824
Front Load Paid by Investors					
Captive	% Inflow	2.40%	1.80%	2.20%	25,807
Unaffiliated	% Inflow	2.77%	2.46%	2.08%	123,824
Captive: Max. Front Load (Morningstar)	% Inflow	4.57%	5.00%	1.41%	25,807
Unaffiliated: Max. Front Load (Morningstar)	% Inflow	4.64%	4.75%	1.44%	123,824
For Those Funds with 12b1 Plans					
12b1 Payments	% TNA	0.30%	0.26%	0.24%	199,618
Funds with Defensive 12b1 Plans	% Funds	11.69%	0	32.13%	199,094
Defensive 12b1 Payments	%TNA	0.17%	0.07%	0.22%	23,274

Panel B: Correlation between N-SAR Actual Loads and Morningstar Maximum Loads

	Front Load	Back Load
Unconditional	0.47	0.44
Positive load	0.14	0.38
Positive load and >80% in A Class	0.38	
Positive load and >80% in B Class		0.33

Baseline Payment Regressions

This table gives the regression estimates of broker payments as a function of fund and family characteristics and other variables. The dependent variable for the panel regression 1 is *Load Paid to Broker*, calculated as the front load dollars paid to both captive and unaffiliated brokers, Q32 + Q33, divided by the inflows subject to a load, Q28h. Regressions 2 and 3 examine *Revenue Sharing* by the fund with the broker. Regression 2 is a probit of whether the fund has a revenue-sharing program as proxied by unreimbursed 12b-1 payments to brokers (Q44) and regression 3 is a panel regression of the dollar amount of revenue sharing (Q44) divided by fund total net assets (TNA). Marginal effects of the probit, *MrgEfx*, measure how a change in the independent variables affects the probability of revenue sharing. While regression 2 includes all funds that charge a 12b-1 fee, regression 3 is limited to those funds that have nonzero revenue-sharing payments. The independent variables are defined in Appendix A. The 38 Morningstar categories are grouped into six general categories with specialty funds omitted for identification and the other fund categories listed as the first five dummies in each regression. The table gives coefficient estimates, t-statistics, adjusted R^2s, and the number of fund-month observations for all three regressions and marginal effects for the probit regression. Panel B gives the p-value for a test of the difference in coefficients. Yearly fixed effects are included and the reported t-statistics use robust standard errors as described in Huber (1967) and White (1980) and cluster by fund.

Panel A: Regressions

	1 Load Paid to Broker		2 Revenue Sharing Indicator			3 Revenue Sharing	
	Coef	t-stat	MrgEfx	Coef	t-stat	Coef	t-stat
Equity	0.0027	4.02	−0.026	−0.196	−1.63	6.76*E*–05	0.29
International	0.0031	4.27	−0.026	−0.212	−1.46	5.81*E*–05	0.21
Fixed Income	0.0031	4.55	−0.024	−0.188	−1.34	9.55*E*–05	0.38
Balanced	0.0028	3.95	−0.029	−0.242	−1.75	2.75*E*–04	1.08
Muni	0.0036	5.42	−0.011	−0.081	−0.56	−2.71*E*–04	−1.03
Front Load	0.5561	30.01					
Front Load × Captive	0.0103	0.51					
Front Load × Unaffil	0.2002	11.73					
Lag Ranked Returns	−4.65*E*–04	−2.69	0.005	0.039	0.82	2.27*E*–04	2.07
Log Family Size	−4.65*E*–04	−6.55	0.019	0.141	6.79	1.34*E*–05	0.41
Log Fund Size	−8.17*E*–05	−1.36	−0.002	−0.016	−0.90	−2.53*E*–04	−6.89
Category Inflows	−0.0375	−7.68	−0.055	−0.400	−0.39	0.0014	0.56
Category Redemptions	−0.0160	−3.50	0.092	0.664	0.56	−2.28*E*–03	−0.74
Family Average Load Paid to Broker	0.2028	15.86					
Family Average Revenue Sharing						0.6543	12.65
Proportion Class A	0.0018	5.48	0.001	0.004	0.04	9.37*E*–04	4.70
Proportion Class B			−0.018	−0.129	−0.64	5.25*E*–04	1.01
Proportion Class C			0.122	0.886	4.13	0.0029	4.14
Proportion Retirement			−0.069	−0.499	−1.64	0.0014	2.82
Proportion Institutional			−0.126	−0.911	−7.17	0.0000	0.15
Captive	0.0019	7.33	−0.081	−1.049	−8.49	0.0013	3.12
Unaffil	−5.86*E*–04	−2.88	−0.008	−0.056	−0.88	−4.47*E*–04	−3.36
Constant	0.0042	3.35		−3.015	−8.73	0.0030	4.95
Year fixed effects	Yes			Yes		Yes	
Adjusted R^2	87.17%			10.46%		39.53%	
Observations	163,347			178,693		21,781	

Panel B: Tests

	F-test	p-value
Front Load × captive = Front Load × Unaffil	189.08	0.001

7.2 Table II: Baseline payment regressions

Read:

- Regression 1 predicts load paid to brokers. Front load has a strong positive relation with load paid to brokers.

- The front-load coefficient is 0.5561, meaning higher investor loads are partly passed to brokers.
- The interaction with unaffiliated brokers is positive and large, indicating unaffiliated brokers receive more of the load.
- The revenue-sharing probit shows that revenue-sharing plans are less likely when funds rely on captive brokers or institutional money.
- The revenue-sharing regression shows higher revenue sharing for retail share classes, especially C shares.
- Family-average load and family-average revenue sharing explain substantial cross-sectional variation.

Economic message: Table II constructs the central residual variables: excess load paid to brokers and excess revenue sharing. These residuals represent broker payments beyond normal fund and family characteristics.

7.3 Table III: Flows and excess load paid to broker

Read:

- Excess load paid to brokers positively predicts inflows subject to load.
- In the base specification, the coefficient on excess load paid to brokers is 0.0372 with a t-statistic of 2.47.
- When split by channel, the effect is negative or weak for captive brokers but positive for unaffiliated brokers.
- The coefficient on excess load paid to brokers times unaffiliated is 0.0952 in specification 2 and 0.1320 in specification 3.
- Front loads reduce inflows, and the negative effect is particularly strong for unaffiliated brokers.
- Past performance matters: high lagged ranked returns attract inflows, and the convexity of performance-flow sensitivity is stronger for unaffiliated brokers.
- Redemptions also respond to lagged redemptions and fund characteristics, but the sales-incentive effect is most relevant for inflows.

Economic message: Broker compensation affects consumer purchases. The effect is strongest where theory predicts it should be strongest: unaffiliated brokers who can steer investors across many fund families.

Table III

Flows and Excess Load Paid to Broker

This table provides regression estimates of inflows and redemptions on residual broker payments from Table II and other fund and family characteristics. The dependent variable for regressions 1 through 3 is monthly fund inflows subject to a load and for regressions 4 through 6 is monthly fund redemptions as a percent of fund total net assets. In regressions 1 and 4, the sample consists of all fund-month observations where the fund reports a nonzero front load collected from investors. Regressions 2 and 5 require that loads are paid solely to unaffiliated or captive brokers, but not both. Regressions 3 and 6 require that the fund family consist of 10 or more funds. The independent variable of interest is *Excess Load Paid to Broker*, the residuals from regression 1 of Table II . The performance measure, *Lag Ranked Returns*, is the fractional rank of the fund's return (between zero and one) from the end of month $t-13$ to the end of month $t-1$ among funds in the same investment objective category. The low and high *Lag Ranked Returns* are calculated as *Lag Ranked Returns Low* = min(0.5, *Lag Ranked Returns*) and *Lag Ranked Returns High* = *Lag Ranked Returns* – *Lag Ranked Returns Low* to create a piecewise linear specification with a kink at the 50th percentile of ranked performance. Lagged inflows subject to a load and redemptions are six-month lags of the dependent variables. All other independent variables are defined in Appendix A. The table gives coefficient estimates, t-statistics, adjusted R^2s, and the number of fund-month observations for all regressions. Panel B gives p-values for tests of the difference in coefficients. Yearly fixed effects are included in the regression and the reported t-statistics use robust standard errors as described in Huber (1967) and White (1980) and cluster by fund.

Panel A: Regressions

	Inflows Subject to Load						Redemptions					
	1		2		3		4		5		6	
	Coef	t-stat	Coef	t-stat	Coef	t-stat	Coef	t-stat	Coef	t-stat	Coef	t-stat
Lagged Inflows Subject to Load	0.4495	34.48	0.4318	30.06	0.4322	27.72						
Lagged Redemptions							0.3666	21.87	0.3576	21.80	0.3466	19.53
Excess Load Paid to Brokers	0.0372	2.47					0.0171	0.52				
Excess Load Paid to Brokers × captive			−0.0899	−2.13	−0.1734	−3.46			−0.1650	−2.67	−0.3059	−4.15
Excess Load Paid to Brokers × Unaffil			0.0952	3.11	0.1320	3.49			−0.0262	−0.65	−0.1214	−2.41
Front Load	−0.1577	−23.44					−0.0523	−5.36				
Front Load × Captive			−0.1228	−7.20	−0.1338	−6.62			−0.0589	−2.81	−0.0277	−1.27
Front Load × Unaffil			−0.1794	−19.89	−0.1866	−19.12			−0.0682	−6.19	−0.0456	−3.89
Redemption Fee	−0.0009	−2.82	−0.0006	−1.72	−0.0013	−3.80	−0.0010	−2.34	−0.0009	−1.85	−0.0006	−1.25
Lag Ranked Returns Low	−0.0003	−0.41					−0.0143	−10.51				
Lag Ranked Returns High	0.0114	11.68					−0.0010	−0.86				
Lag Ranked Returns Low × captive			0.0041	2.21	0.0058	2.77			−0.0098	−3.81	−0.0082	−3.36
Lag Ranked Returns Low × Unaffil			−0.0016	−1.46	−0.0013	−1.12			−0.0150	−8.63	−0.0142	−7.78
Lag Ranked Returns High × Captive			0.0046	2.05	0.0041	1.59			−0.0006	−0.25	0.0008	0.31
Lag Ranked Returns High × Unaffil			0.0138	10.67	0.0132	9.53			−0.0010	−0.76	−0.0007	−0.45
Log Family Size	0.0007	8.15	0.0009	8.56	0.0013	9.52	0.0007	4.22	0.0008	5.77	0.0005	2.89
Log Fund Size	−0.0011	−9.49	−0.0012	−9.27	−0.0012	−8.48	−0.0009	−6.39	−0.0009	−5.91	−0.0006	−3.60
Expenses	0.1886	5.33	0.1606	4.03	0.1365	3.04	0.3993	7.61	0.3325	6.75	0.2867	5.33
Category Inflows	0.1879	10.23	0.2184	8.79	0.2131	7.76						
Category Redemptions							0.6919	21.52	0.7201	21.57	0.7262	20.28
Proportion Class A	0.0044	9.22	0.0039	7.21	0.0038	6.08	−0.0029	−3.73	−0.0041	−5.11	−0.0039	−4.33
Captive			−0.0042	−4.47	−0.0078	−4.32			−0.0048	−3.01	−0.0094	−4.22
Family Inflows × Captive					0.1271	2.75					0.1292	2.05
Family Inflows × Unaffil					0.0873	6.07					0.0417	1.95
Family Redemptions × Captive					0.0890	2.17					0.3619	6.98
Family Redemptions × Unaffil					−0.0090	−0.55					0.2785	9.38
Constant	0.0093	1.66	0.0014	0.74	0.0020	0.27	0.0060	1.77	0.0039	1.44	0.0068	0.72
Year Fixed Effects	Yes		Yes		Yes		Yes		Yes		Yes	
Adjusted R^2	35.76%		36.38%		36.81%		30.43%		30.55%		33.44%	
Observations	126,904		98,063		83,781		126,979		98,121		83,839	

Panel B: Tests

	2		3		4		5	
	F-test	p-value	F-test	p-value	F-test	p-value	F-test	p-value
Excess Load Paid to Broker (Captive = Unaffil)	11.82	0.001	22.14	0.001	3.50	0.062	4.29	0.039
Front Load (Captive = Unaffil)	8.54	0.004	5.65	0.018	0.15	0.699	0.52	0.473
Lag Ranked Return Low (Captive = Unaffil)	6.99	0.008	8.73	0.003	2.68	0.102	3.81	0.051
Lag Ranked Return High (Captive = Unaffil)	12.33	0.001	9.63	0.002	0.03	0.873	0.24	0.621
Lag Ranked Return High–Low (Captive)	0.02	0.889	0.16	0.686	4.93	0.027	4.32	0.038
Lag Ranked Return High–Low (Unaffil)	53.73	0.000	40.52	0.000	24.46	0.000	20.42	0.000
Family Inflows (Captive = Unaffil)			0.68	0.410			1.73	0.188
Family Redemptions (Captive = Unaffil)			5.06	0.025			2.13	0.145

7.4 Table IV: Flows and excess revenue sharing

Read:

- Excess revenue sharing positively predicts inflows in the revenue-sharing sample.
- The coefficient on excess revenue sharing is 1.1765 in the first inflow regression and 1.1199 in the second.
- Revenue sharing is weaker for redemptions and not consistently significant there.
- Lagged ranked returns have the usual performance-flow effect: low-ranked funds receive less inflow, high-ranked funds receive more inflow.
- Family inflows and category flows also strongly predict fund inflows.

Economic message: Revenue sharing also influences where consumer money goes. However, its welfare implications differ from load sharing because revenue sharing is linked to continuing assets under management.

Flows and Excess Revenue Sharing

This table gives the regression estimates of inflows and redemptions on excess broker payments from Table II and other fund and family characteristics. The dependent variable for regressions 1 and 2 is monthly fund inflows as a percent of fund total net assets (TNA). The dependent variable for regressions 3 and 4 is monthly fund redemptions as a percent of fund TNA. In regressions 1 and 3, the sample consists of all fund-month observations where the fund reports unreimbursed 12b-1 expenses/revenue-sharing payments (Q44). In regressions 2 and 4 a requirement that the fund family have 10 or more funds is added. The independent variable of interest is *Excess Revenue Sharing*, the residuals from regression 3 in Table II. The performance measure, *Lag Ranked Returns*, is the fractional rank of the fund's return (between zero and one) from the end of month $t-13$ to the end of month $t-1$ among funds in the same investment objective category. The low and high *Lag Ranked Returns* are calculated as *Lag Ranked Returns Low* = min(0.5, *Lag Ranked Returns*) and *Lag Ranked Returns High* = *Lag Ranked Returns* – *Lag Ranked Returns Low* to create a piecewise linear specification with a kink at the 50th percentile of ranked performance. Lagged inflows and redemptions are six-month lags of the dependent variable. All other independent variables are defined in Appendix A. The table gives coefficient estimates, t-statistics, adjusted R^2s, and the number of fund-month observations. Yearly fixed effects are included in the regression and the reported t-statistics use robust standard errors as described in Huber (1967) and White (1980) and cluster by fund.

	Inflows				Redemptions			
	1		2		3		4	
	Coef	t-stat	Coef	t-stat	Coef	t-stat	Coef	t-stat
Lagged Inflows	0.4097	14.20	0.3733	17.72				
Lagged Redemptions					0.3783	6.43	0.3234	8.02
Excess Revenue Sharing	1.1765	3.07	1.1199	2.87	0.3802	1.21	0.1986	0.73
Front Load	−0.0987	−2.28	−0.1240	−4.51	−0.0599	−1.48	−0.0605	−3.32
Back Load	0.0362	1.37	0.0361	1.28	−0.0071	−1.05	−0.0070	−1.19
Redemption Fee	0.0010	0.66	0.0008	0.56	−0.0019	−1.82	−0.0014	−1.40
Lag Ranked Returns Low	−0.0014	−0.41	−0.0005	−0.14	−0.0162	−5.04	−0.0135	−4.37
Lag Ranked Returns High	0.0309	6.74	0.0317	7.86	−0.0001	−0.04	0.0026	0.88
Log Family Size	0.0011	1.49	0.0030	4.54	−0.0003	−0.41	0.0004	0.82
Log Fund Size	−0.0033	−6.19	−0.0039	−8.24	−0.0011	−3.03	−0.0011	−3.68
Expenses	0.0372	0.17	0.1585	0.66	0.3687	1.67	0.7210	3.70
Proportion Class A	−0.0002	−0.08	−0.0003	−0.09	−0.0005	−0.22	−0.0031	−1.21
Proportion Class B	−0.0065	−1.18	−0.0007	−0.10	−0.0083	−1.87	−0.0132	−2.20
Proportion Class C	−0.0008	−0.11	−0.0054	−0.69	0.0050	0.89	−0.0018	−0.28
Proportion Retirement	0.0216	2.18	0.0294	2.89	0.0115	1.47	0.0130	1.53
Proportion Institutional	−0.0012	−0.29	−0.0010	−0.21	−0.0009	−0.32	−0.0056	−1.77
Category Inflows	0.6479	8.25	0.7544	10.63				
Category Redemptions					0.6603	8.82	0.7265	10.93
Family Inflows			0.2225	3.57			0.0737	1.36
Family Redemptions			0.0500	0.73			0.1974	2.91
Constant	0.0580	1.68	−0.0172	−1.34	0.0524	2.36	0.0051	0.49
Year Fixed Effects	Yes		Yes		Yes		Yes	
Adjusted R^2	39.34%		44.32%		31.21%		36.30%	
Observations	17.479		14.921		17.479		14.921	

Future Returns and Excess Load Paid to Broker

This table gives regression estimates of 12-month forward-looking net excess returns as a function of broker payments and other fund and family characteristics. The dependent variable is the compounded monthly fund returns net of expenses from $t+1$ to $t+12$ in excess of the compounded Morningstar investment objective category net returns over the same time period. In regressions 1, 2, and 3 the sample consists of all fund-month observations where the fund reports a nonzero front load collected from investors. Regression 4 is the same sample but with the added requirement that loads are paid solely to unaffiliated or captive brokers. The independent variable of interest is *Excess Load Paid to Brokers*, the residuals from regression 1 of Table II. Also, in specification 3 *New Inflows* are separated into *Inflows Subject to a Load* and *Inflows Not Subject to a Load*, which along with all other independent variables are defined in Appendix A. In Panel A, the table lists coefficient estimates, t-statistics, adjusted R^2s, and the number of fund-month observations for all four regressions. Panel B gives p-values for tests of the difference in coefficients. Both year and Morningstar investment objective fixed effects (38 categories) are included in the regression and the reported t-statistics use robust standard errors as described in Huber (1967) and White (1980) and cluster by fund.

Panel A: Regressions

	Excess Returns Months +1 to +12							
	1		2		3		4	
	Coef	t-stat	Coef	t-stat	Coef	t-stat	Coef	t-stat
Inflows	0.0526	3.10	0.0014	0.09			0.0589	2.96
Inflows Subject to Load					0.0045	0.16		
Inflows Not Subject to Load					0.0758	3.45		
Redemptions	−0.1287	−6.61	−0.0731	−3.97	−0.1290	−6.59	−0.1269	−5.28
Log Family Size	0.0025	5.10	0.0023	5.01	0.0026	5.12	0.0023	4.04
Log Fund Size	−0.0038	−6.48	−0.0041	−7.42	−0.0038	−6.51	−0.0030	−5.02
Category Inflows	−0.0083	−0.16	0.0122	0.24	−0.0111	−0.21	0.0336	0.48
Category Redemptions	0.1579	2.76	0.1224	2.16	0.1574	2.75	0.1935	2.78
Excess Load Paid to Broker	−0.3441	−2.84	−0.3523	−3.01	−0.3338	−2.77		
Excess Load Paid to Broker × Captive							−0.1448	−0.88
Excess Load Paid to Broker × Unaffil							−0.4972	−2.48
Lag Ranked Returns			0.0376	15.80				
Captive							−0.0011	−0.60
Constant	0.0312	3.25	0.0184	2.03	0.0314	3.27	0.0192	1.66
Year Fixed Effects	Yes		Yes		Yes		Yes	
Objective Category Fixed Effects	Yes		Yes		Yes		Yes	
Adjusted R^2	1.54%		3.13%		1.56%		1.48%	
Observations	146,461		146,461		146,461		113,153	

Panel B: Tests

	F-test	p-value	F-test	p-value
New Inflows (Subj Load = Not Subj Load)	3.68	0.0551		
Excess Load Paid to Broker (Captive = Unaffil)			1.96	0.1616

7.5 Table V: Future returns and excess load paid to broker

Read:

- Excess load paid to brokers predicts lower future 12-month excess returns.
- In specification 1, the coefficient on excess load paid to brokers is −0.3441 with a t-statistic of −2.84.
- The estimate implies that 1% more paid to brokers predicts about 0.34% lower future performance over the next year.

- Flows not subject to load predict good future performance, while flows subject to load do not.
- The channel split shows that excess load paid to unaffiliated brokers has a significant negative coefficient, around −0.4972.
- The captive-channel interaction is not statistically significant.

Economic message: The same payments that attract inflows through brokers are associated with worse subsequent performance. This is the paper's strongest welfare result.

7.6 Table VI: Future returns and excess revenue sharing

Read:

- Without controlling for load sharing, excess revenue sharing enters negatively in specification 1.
- Once excess load paid to brokers is included, the revenue-sharing coefficient is no longer significant.
- Excess load paid to brokers remains negative and significant in specifications 2 and 3.
- Lagged ranked returns positively predict future performance in specification 3.

Economic message: The performance cost is tied more clearly to upfront load sharing than to revenue sharing. This is consistent with the interpretation that load sharing rewards sales, while revenue sharing gives brokers continuing exposure to fund assets.

Future Returns and Excess Revenue Sharing

This table gives regression estimates of 12-month forward-looking net excess returns as a function of broker payments and other fund and family characteristics. The dependent variable is the compounded monthly fund returns net of expenses from $t+1$ to $t+12$ in excess of the compounded Morningstar investment objective category net returns over the same time period. In regression 1 the sample consists of all fund-month observations where the fund reports a nonzero revenue sharing or unreimbursed 12b-1 payment. Regressions 2 and 3 require that the fund have a nonmissing excess load payment to the broker. The independent variables *Excess Load Paid to Broker* and *Excess Revenue Sharing* are the residuals from regressions 1 and 3, respectively, of Table II. All other independent variables are defined in Appendix A. The table gives coefficient estimates, t-statistics, adjusted R^2s, and the number of fund-month observations. Both year and Morningstar investment objective fixed effects (38 categories) are included in the regression and the reported t-statistics use robust standard errors as described in Huber (1967) and White (1980) and cluster by fund.

	Excess Returns Months +1 to +12					
	1		2		3	
	Coef	t-stat	Coef	t-stat	Coef	t-stat
Inflows	0.0731	1.66	0.0314	0.61	−0.0304	−0.61
Redemptions	−0.0710	−1.49	−0.0292	−0.52	0.0425	0.75
Log Family Size	−0.0006	−0.44	−0.0003	−0.21	−0.0004	−0.32
Log Fund Size	−0.0027	−1.62	−0.0032	−1.69	−0.0038	−2.07
Category Inflows	−0.1398	−0.88	−0.1237	−0.71	−0.0772	−0.45
Category Redemptions	0.1626	1.02	0.2259	1.31	0.1647	0.96
Excess Revenue Sharing	−2.4638	−2.50	−1.1230	−1.23	−1.1464	−1.29
Excess Load Paid to Broker			−1.1236	−2.52	−1.0547	−2.54
Lag Ranked Returns					0.0387	5.19
Constant	0.0819	2.36	0.0484	0.94	0.0334	0.67
Year Fixed Effects	Yes		Yes		Yes	
Objective Category Fixed Effects	Yes		Yes		Yes	
Adjusted R^2	3.20%		2.96%		4.46%	
Observations	19,541		15,346		15,346	

7.7 Appendix A: Variable definitions

Read:

- Appendix A defines the N-SAR variables used to construct captive, unaffiliated, brokered, inflows, redemptions, loads, and revenue-sharing measures.
- It also defines Morningstar and asset-weighted variables such as lagged ranked return, fund age, fund size, and objective category.

Economic message: The appendix is important because many variables are not standard CRSP or Morningstar measures; they are constructed from regulatory disclosure fields.

Appendix A. Variable Definitions

Table AI
Variables Using N-SAR Data

Variable	Definition
Captive	= 1 if fund i received loads through captive brokers (Q33) but not unaffiliated brokers (Q32) in *NSAR(i,t)*, = 0 otherwise.
Unaffil	= 1 if fund i received loads through unaffiliated brokers (Q32) but not captive brokers (Q33) in *NSAR(i,t)*, = 0 otherwise.
Brokered	= 1 if fund i received loads through unaffiliated brokers (Q32) or captive brokers (Q33), = 0 otherwise.
Revenue Sharing Indicator	= 1 if fund i indicates that it participates in a defensive 12b-1 plan (Q44), = 0 otherwise.
Revenue Sharing	Amount of payments made pursuant to a defensive 12b-1 plan (Q44) divided by average net assets (Q75). Because we use the semiannual data, this number is multiplied by two to annualize.
Inflows	$\frac{\text{Investments in fund } i \text{ in month } t \text{ from Q28a to Q28f of the NSAR}}{\text{Total net assets in fund } i \text{ at the end of month } t \text{ from Morningstar}}$
Proportion Inflows Subject to Load	$\frac{\text{Inflows subject to Load (Q28h)}}{\text{Total new inflows (Q28g1 + Q28g2 + Q28g3)}}$
Inflows Subject to Load	New Inflows × Proportion Inflows Subject to Load
Inflows Not Subject to Load	New Inflows × (1−Proportion Inflows Subject to Load)
Redemptions	$\frac{\text{Redemptions in fund } i \text{ in month } t \text{ from Q28a to Q28f of the NSAR}}{\text{Total net assets in fund } i \text{ at the end of month } t \text{ from Morningstar}}$
Front Load	$\frac{\text{Front loads paid by consumers from Q30a of the NSAR}}{\text{Inflows subject to a load from Q28h of the NSAR}}$

Appendix A: N-SAR variables

Table AII
Variables from Morningstar Asset-Weighted across All Share Classes of the Same Fund

Variable	Definition
Lag Ranked Return	Fractional rank of the fund's net return (between 0 and 1) from the end of month $t-13$ to the end of month $t-1$ among funds in the same investment objective category.
Log Fund Size	Log of total assets (in thousands) of fund i at the end of month t.
Fund Age	Current age of the fund in years since the fund's inception.
Index Fund	Takes the value one if the fund is an index fund and zero otherwise.
Log Family Size	Log of total assets (in thousands) of all funds in i's family with available N-SAR data as of month t.
Proportion Share Class (A, B, C, etc.)	Portion of assets held in five different share class categories ranging from 0 to 1: A, B, C, Institutional and Retirement. Share class designations are determined from the Morningstar's "Share Class Type" variable.
Objective Category	Morningstar sorts funds into 38 categories. We group these categories into equity, fixed income, international, municipal, specialty, and balanced.
Excess Returns	Compounded monthly fund returns net of expenses over a 12-month horizon (+1 to +12 months) in excess of the compounded Morningstar investment objective category net returns over the same time period (reported in decimals).

Table AIII
Family-Level Variables

Variable	Definition

Appendix A: Morningstar and family variables

8 Short summary

- The paper studies whether mutual fund flows respond to broker compensation.
- It uses SEC N-SAR filings merged with Morningstar data for U.S. funds from 1993 to 2009.
- N-SAR data reveal actual front loads, payments to captive and unaffiliated brokers, revenue-sharing proxies, and separate inflow and outflow components.
- Funds that pay brokers more receive more inflows, especially through unaffiliated brokers.
- Higher total loads reduce inflows, especially when investors buy through unaffiliated brokers.
- Revenue sharing also predicts inflows in the revenue-sharing sample.
- Load payments to brokers predict lower future fund performance.
- The negative performance effect is strongest for load payments to unaffiliated brokers.
- Revenue sharing does not predict poor performance once load sharing is controlled for.
- Captive brokerage helps fund families recapture redemptions elsewhere in the same family.
- The paper provides evidence that broker compensation can distort consumer investment decisions and reduce investor welfare.

9 Conclusion

9.1 Core conclusion

Christoffersen, Evans, and Musto show that broker compensation affects mutual fund flows and investor outcomes. Consumers' fund flows do not simply maximize fund performance or minimize investor costs. They also respond to the incentives created by load sharing and revenue sharing between fund sponsors and brokers.

9.2 Economic intuition

When a fund pays more of the load to a broker, the broker has a stronger incentive to recommend that fund. This incentive is especially powerful for unaffiliated brokers because they sell funds from many families and can steer investors across alternatives. But load sharing rewards the sale rather than future performance, so the induced inflows can be harmful to consumers.

9.3 Incremental contribution

The paper's incremental contribution is to use regulatory N-SAR disclosures to open the black box of broker compensation. Earlier studies could observe loads and net flows, but not the actual split of loads between fund sponsors and brokers, not the distinction between captive and unaffiliated channels, and not the separation of inflows from redemptions. This paper uses those distinctions to show that broker incentives matter for both sales and welfare.

9.4 Policy implication

The evidence supports regulatory concern over broker conflicts of interest. It also suggests that the details of compensation matter: upfront load sharing is more clearly linked to poor future performance than revenue sharing. Better disclosure of the cross-section of broker payments and the form of those payments could help consumers evaluate broker recommendations.